

Advanced Growth Econometrics

Instructor:	Prof. Elena Rostova	Term:	Summer/Fall 2026
Email:	erostova@univ.edu	Dates:	May 27 – Sept 15, 2026
Office:	Heroy Hall 214	Office Hours:	Tue / Thu 10:00 AM – 12:00 PM

COURSE DESCRIPTION

Application of statistical frameworks to high-conviction growth markets and alternative asset classes. Covers multi-variable regression models, time-series forecasting, and the quantification of tracking error and volatility.

GRADING POLICY & WEIGHTS

Assessment Activity	Weight
Participation (Qwickly Check-ins)	10%
Data Analysis Homework	20%
Statistical Modeling Labs (R/Python or Excel)	20%
Midterm Examination	25%
Final Econometric Research Paper	25%

COURSE SCHEDULE

Date / Session	Topics, Assignments & Deadlines
May 28 (Thu)	Introduction to Econometric Modeling & Data Integrity
June 4 (Thu)	Module 1: Ordinary Least Squares (OLS) Estimators
June 9 (Tue)	Homework 1 Due: Simple Linear Regression Models
June 11 (Thu)	Module 1: Multiple Regression Mechanics & Multi-collinearity
June 18 (Thu)	Lab 1 Due: Identifying Cross-Sectional Variance
June 25 (Thu)	Module 2: Dummy Variables & Qualitative Regressors
July 2 (Thu)	Module 2: Heteroskedasticity Detection & Robust Standard Errors
July 7 (Tue)	Homework 2 Due: Variance Inflation Factors

Date / Session	Topics, Assignments & Deadlines
July 9 (Thu)	MIDTERM EXAMINATION (Modules 1–2)
July 16 (Thu)	Module 3: Time Series Analysis & Autoregressive Models
July 23 (Thu)	Module 3: Moving Averages & Trend Projections in Market Volatility
July 28 (Tue)	Lab 2 Due: Autoregressive Integrated Moving Average (ARIMA) Models
July 30 (Thu)	Module 4: Panel Data Analysis & Fixed Effects Models
August 6 (Thu)	Module 4: Instrumental Variables & Two-Stage Least Squares
August 13 (Thu)	Homework 3 Due: Endogeneity Corrections
August 20 (Thu)	Module 5: Logistic Regression & Binary Choice Frameworks
August 27 (Thu)	Lab 3 Due: Probability Score Adjustments
September 3 (Thu)	Draft Paper Review and Peer-Led Critique Sessions
September 10 (Thu)	Final Paper Submission & Review
September 15 (Tue)	FINAL RESEARCH PAPER DEADLINE (11:59 PM)